

Memory Circuits & Systems

Exercise 4

DRAM Differential Sense Amplifier

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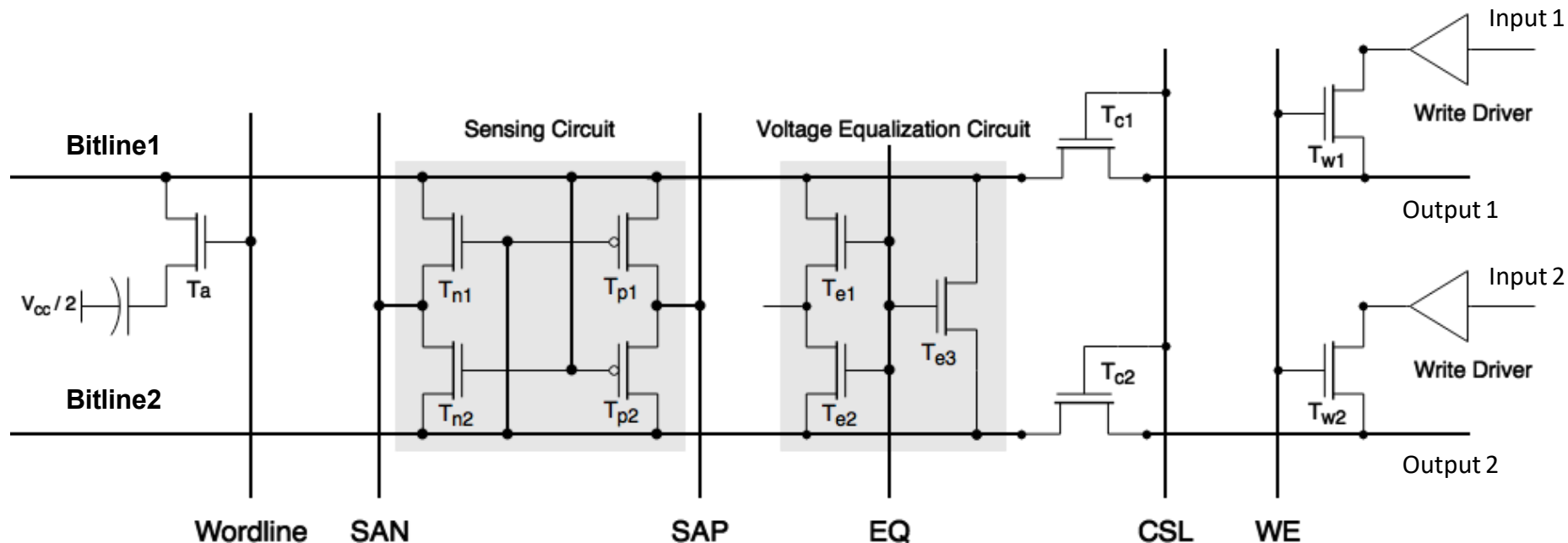


1.DRAM Differential Sensing Amplifier

- Implementation Read & Write operation of DRAM using 7nm model
 - ◆ Input: WE, CSL, EQ, SAP, SAN, WL
 - ◆ Output: Output1 of Bitline1, Output2 of Bitline2
 - ◆ Design a differential sensing amplifier and simulate DRAM Behavior (1T1C cell, $C = 20 \text{ fF}$)
 - ◆ Estimate loading of Bitline1 and Bitline2 (Add an extra C to emulate the parasitic C on BL)
 - $C_{(BL)} : C_{(C)}$ typically in the range $3 : 1 \sim 6 : 1$
 - ◆ Please compare and analyze the relationship of loading of DRAM cell between Bitline1 and Bitline2
 - ◆ Operating Voltage : 0.7V

Differential SA Structure for DRAM

- Architecture of a differential SA for DRAM.



Read/Write mode of DRAM

■ Read mode [25%]

- ◆ Step1: Precharge
- ◆ Step2: Access
- ◆ Step3: Sense
- ◆ Step4: Restore

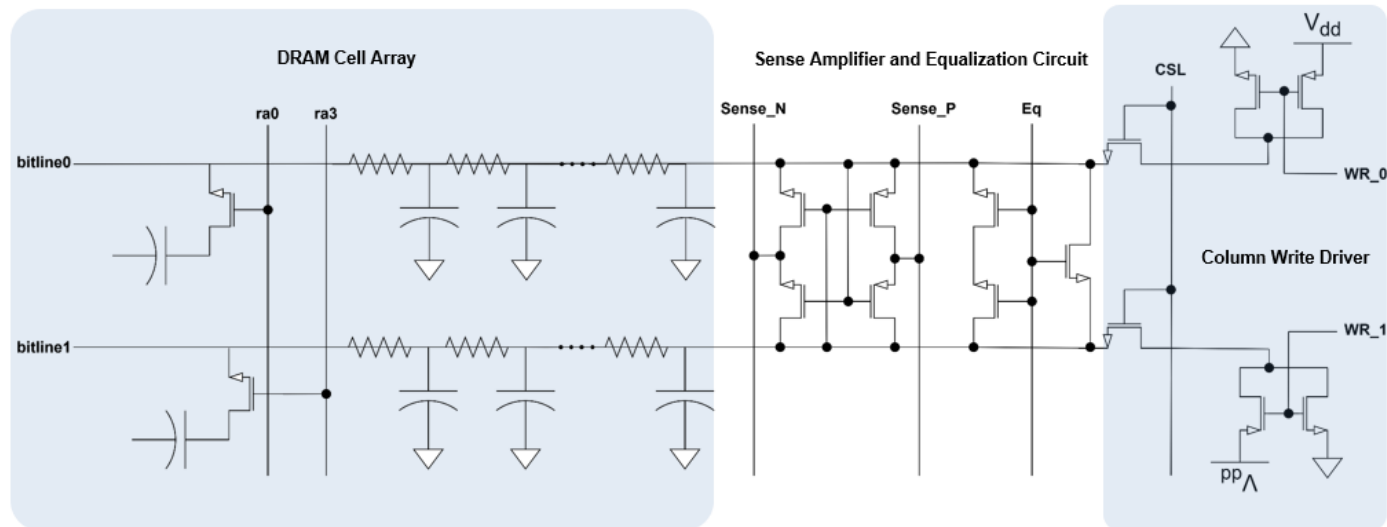
■ Write mode [25%]

- ◆ Step1: Precharge
- ◆ Step2: Access
- ◆ Step3: Sense
- ◆ Step4: Restore
- ◆ Step5: Write Recovery

- Please compare and analyze the relationship of loading of DRAM cell between Bitline1 & Bitline2. [20%]

2. Timing Measurement with predicted model

- Perform transient analysis on the provided 512×512 DRAM array circuit designed in 22 nm technology, and extract the key DRAM timing parameters: tRCD, tRAS, and tRP. [30%]
- ◆ TA will provide DRAM array template includes a parasitic bitline/wordline RC model to emulate the loading of a 32KB DRAM array
- ◆ Implement the Sense Amplifier, the Precharge and Equalization Circuit, and the required control signal generation in your DRAM array design
- ◆ Operating Voltage : 1.2V
- ◆ Wordline Voltage : 2.5V



■ : Provided by TA

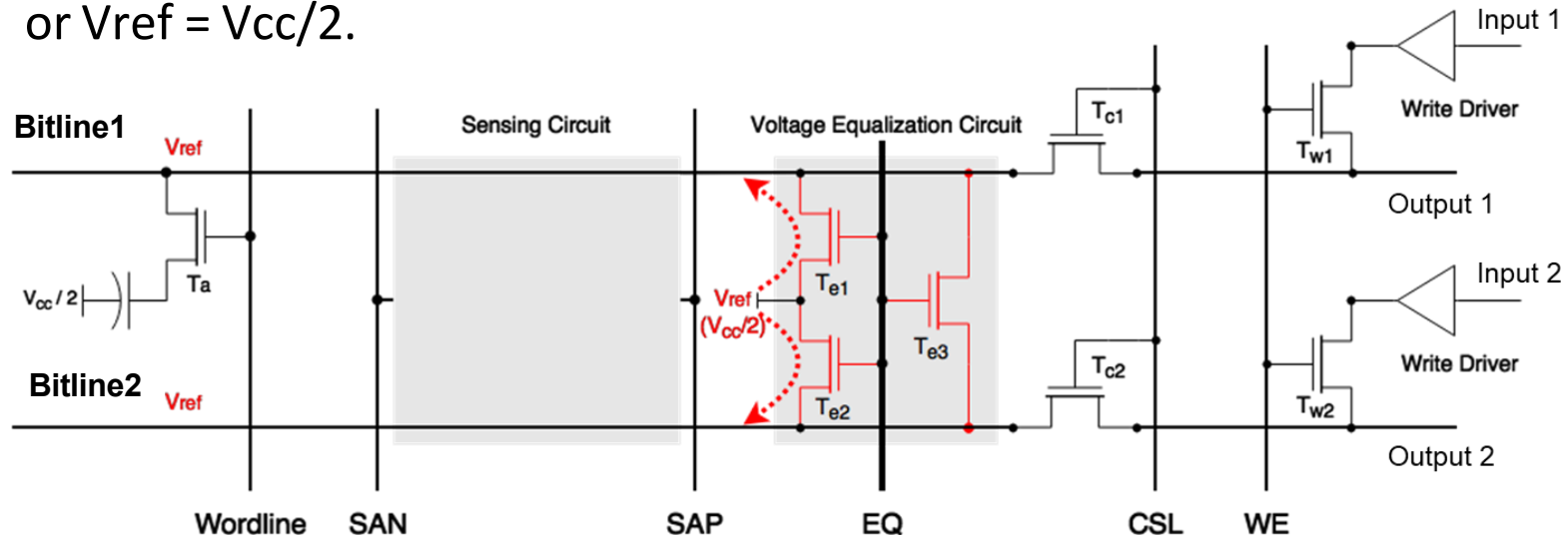
Submission on E3 platform

- Please compress your **report & source codes in a single compressed file (.zip/.rar)** and upload this single file on E3 platform
- ◆ Source code
 - Hspice code of differential SA for DRAM
 - Input pattern (include read & write mode)
- ◆ Report:
 - Waveform: should include the information of each step in Read/Write mode.
 - Comparison & analysis
- ◆ Naming rules of the compressed file
- ◆ Upload file: **Ex4_#ID.zip (including report & source code)**
- Due date: 5/27 P.M. 23:55

Read mode of DRAM

■ Step 1: Precharge

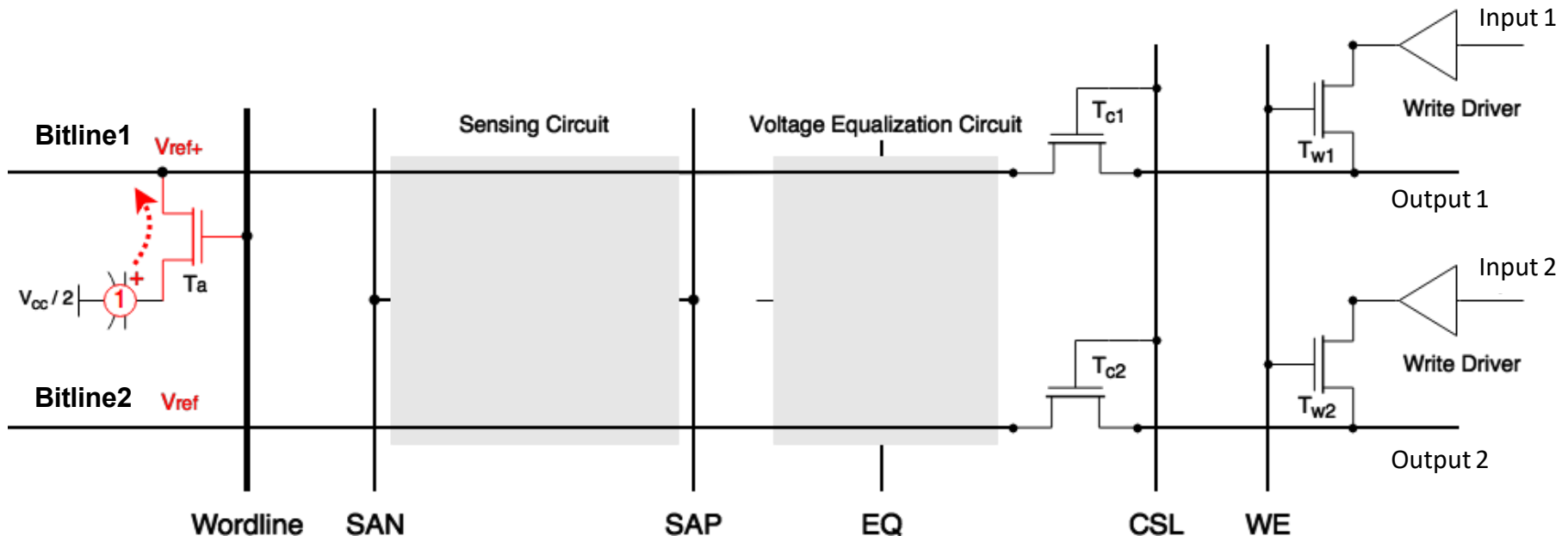
- **Controlling the EQ signal:** The EQ signal is a gate signal used to control the precharge transistors T_{e1} , T_{e2} , and T_{e3} . When the EQ signal is high, these transistors turn on.
- **T_{e1} , T_{e2} , T_{e3} transistors turn on:** When the T_{e1} , T_{e2} , and T_{e3} transistors turn on, they form a path that connects the Bitline1 and Bitline2 to the power supply V_{cc} and the reference voltage V_{ref} .
- **Bitline1 and Bitline2 voltages stabilize at V_{ref} :** Due to the conduction of the T_{e1} , T_{e2} , and T_{e3} transistors, the charges on the Bitline1 and Bitline2 will flow until their voltages stabilize at V_{ref} . V_{ref} is typically half of V_{cc} , or $V_{ref} = V_{cc}/2$.



Read mode of DRAM

■ Step2: Access

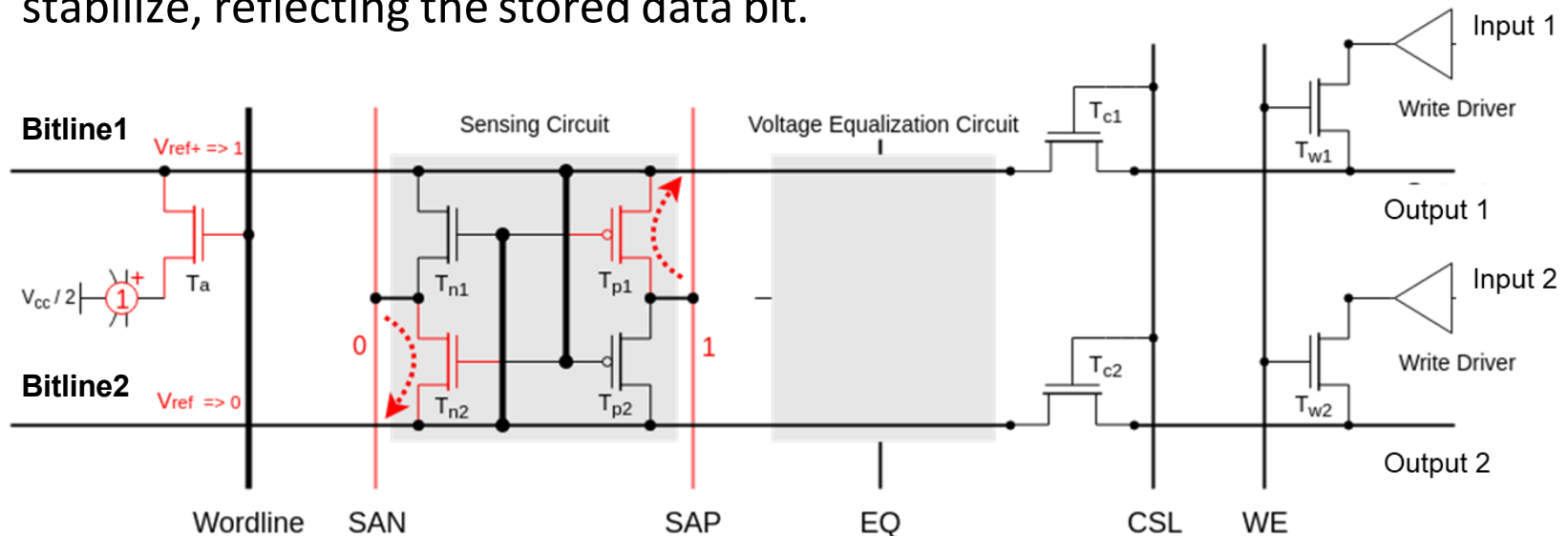
- ◆ **Wordline signal:** The Wordline signal selects the storage cell for data write operation.
- ◆ **Ta transistor:** The Ta transistor connects the Storage Capacitor to the Bitline1.
- ◆ **Data write:** The stored positive charges in the Storage Capacitor flow to the Bitline1, raising its voltage to V_{ref+} .



Read mode of DRAM

Step3: Sense

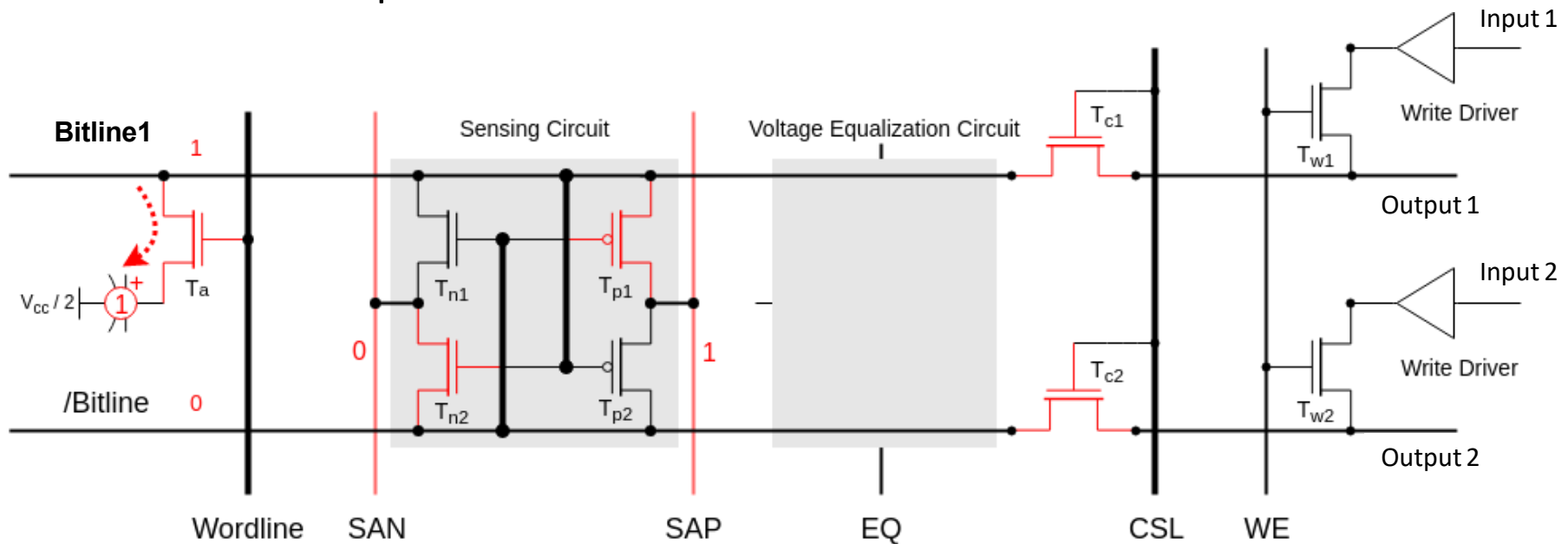
- ◆ **SAN and SAP voltages:** SAN (logic 0) and SAP (V_{cc}) control the sense amplifier.
- ◆ **Tn2 stronger than Tn1, Tp1 stronger than Tp2:** Due to Access stage, Tn2 conducts more than Tn1, and Tp1 conducts more than Tp2.
- ◆ **SAN pulls down Bitline2, SAP pulls up Bitline1:** SAN pulls down Bitline2 faster, and SAP pulls up Bitline1 faster.
- ◆ **Tp1 and Tn2 on, Tp2 and Tn1 off:** Tp1 and Tn2 turn on, Tp2 and Tn1 turn off.
- ◆ **Bitline1 and Bitline2 voltages stabilize:** Bitline1 and Bitline2 voltages stabilize, reflecting the stored data bit.



Read mode of DRAM

Step4: Restore

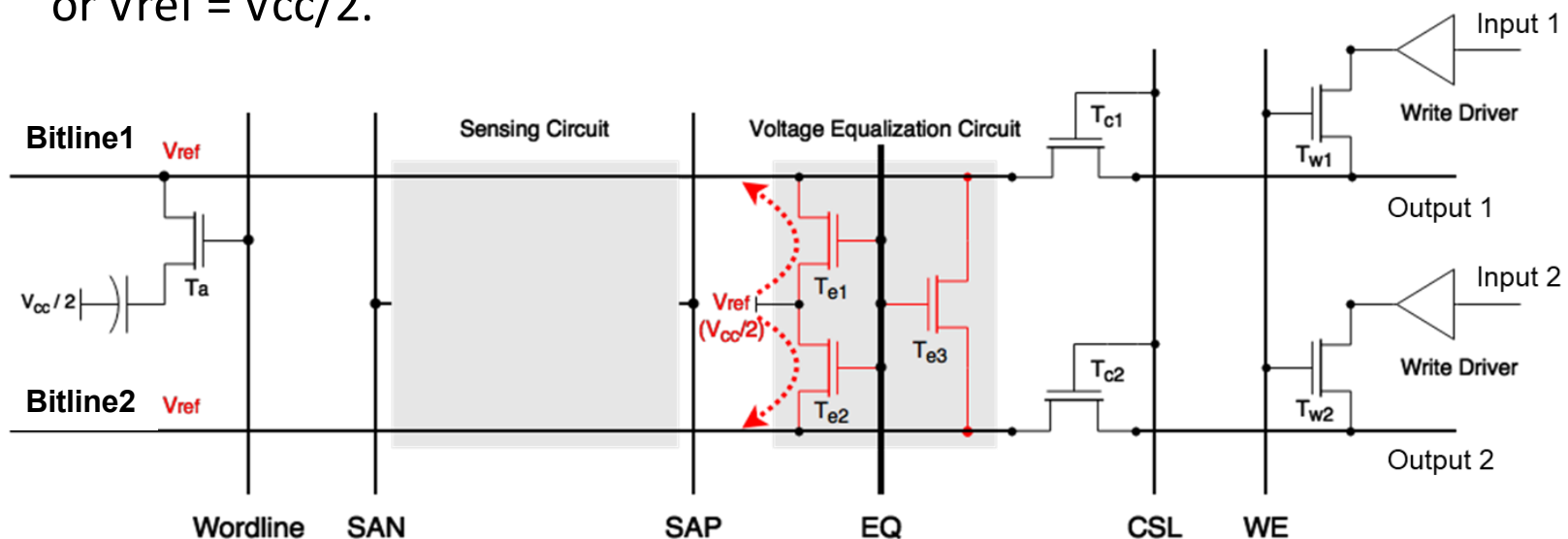
- ◆ **Bitline1 voltage stable:** Bitline1 voltage remains at V_{cc} after the Sense stage.
- ◆ **Bitline1 charges Storage Capacitor:** Bitline1 charges the Storage Capacitor through T_{p1} .
- ◆ **Storage Capacitor charge restored:** The Storage Capacitor's charge is restored to its pre-read state.



Write mode of DRAM

■ Step1: Precharge

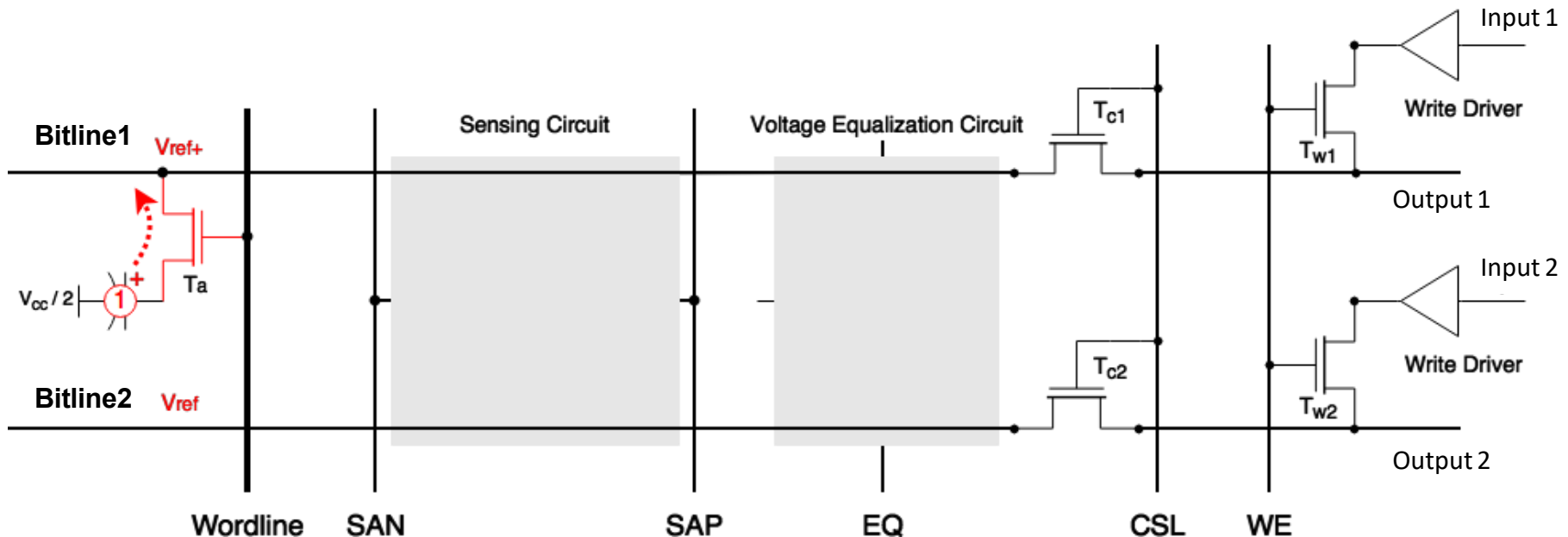
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Write mode of DRAM

Step2: Access

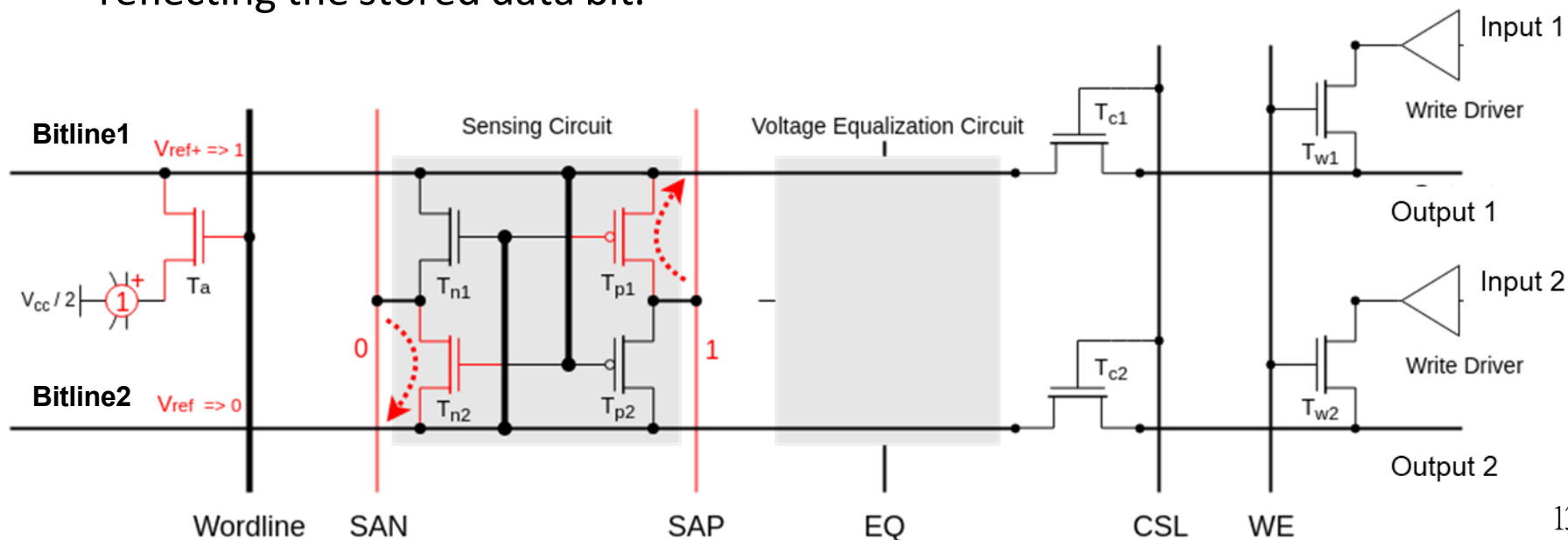
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Write mode of DRAM

Step3: Sense

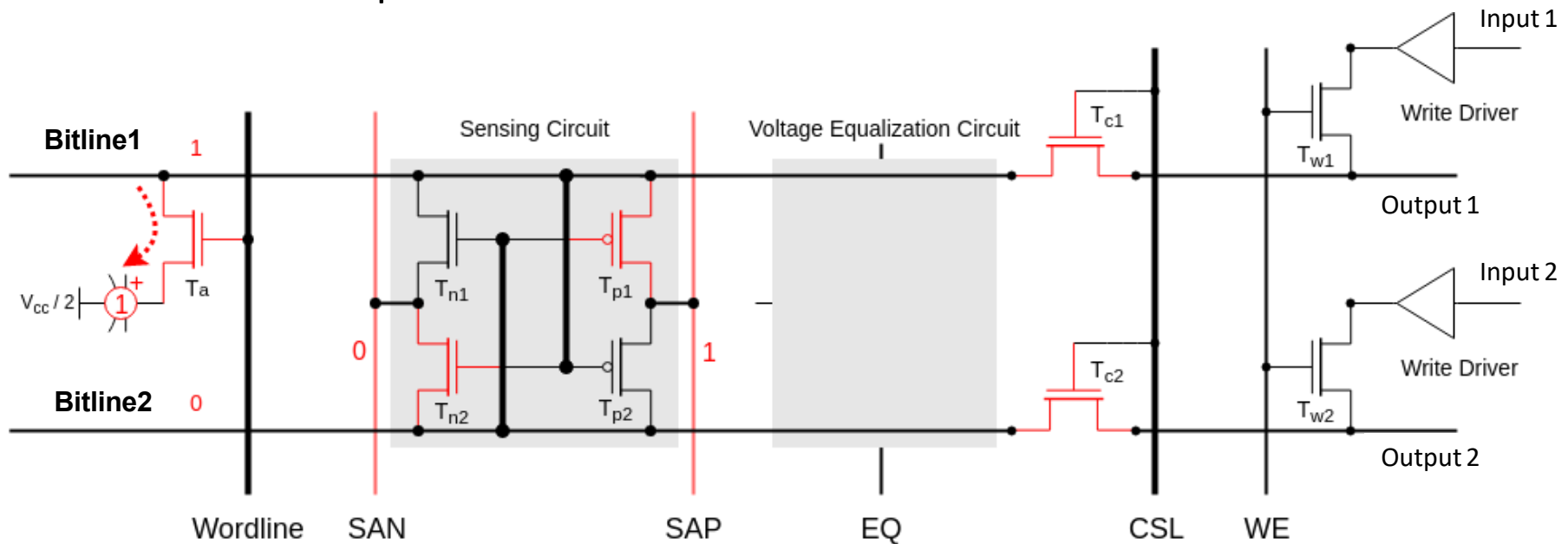
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- ◆ **Bitline1 and Bitline2 voltages stabilize:** Bitline1 and Bitline2 voltages stabilize, reflecting the stored data bit.



Write mode of DRAM

Step4: Restore

- ◆ **Bitline1 voltage stable:** Bitline1 voltage remains at V_{cc} after the Sense stage.
- ◆ **Bitline1 charges Storage Capacitor:** Bitline1 charges the Storage Capacitor through T_{p1} .
- ◆ **Storage Capacitor charge restored:** The Storage Capacitor's charge is restored to its pre-read state.



Write mode of DRAM

Step5: Write Recovery

- ◆ **Maintaining WE signal low:** Prevents further data writing.
- ◆ **Stabilizing Bitline1 and Bitline2 voltages:** Keeps Bitline1 at logic 0 and Bitline2 at logic 1.
- ◆ **Discharging Storage Capacitor:** Gradually discharges the Storage Capacitor to 0.
- ◆ **Cutting off Access Transistor:** Isolates the Storage Capacitor from Bitline1 and Bitline2.
- ◆ **Completion of Write Operation:** Data is stable, and the write operation is complete.

